ORAVONT SYSTEMS

Underwater Acoustic Intelligence

Re-ID User Manual

SKANN Vessel Re-Identification — purpose, technology and operation

Application version: v1.3 (native desktop window; save-dialog exports) · **Model:** `disc5_arcface_8k_ft2_ep003.pth` (frozen) · **Document status:** complete — all screenshots captured and embedded

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Part I — What Re-ID is and why it matters

This part answers one question: what does the Re-ID application do that is not being done today? It describes the operational purpose (§1), the two independent methods the application runs (§2), why the new method is built the way it is (§3), what changes for the operator, the analyst and the unit (§4), and a brief account of the technology (§5). Parts II and III then cover installation and operation step by step. Companion document: *DISC5 Re-ID System Flow*
covers what happens inside, file by file.

1. Purpose

Re-ID answers one operational question: **"Which known vessel does this recording sound like?"** An operator loads a passive-sonar recording (a WAV file), and the application returns a ranked list of candidate vessels from a gallery of previously enrolled recordings, each with a similarity score.

Today that question is answered by an analyst reading LOFAR displays: locating narrowband tonal lines, comparing their frequencies against remembered or recorded line sets, and judging a match. That skill is real and remains valuable — but it is slow, depends on the individual analyst, and degrades under exactly the conditions of operational interest: sea-state noise and closing/opening targets whose tonals are Doppler-shifted.

Re-ID automates the comparison step. It does not replace the analyst's judgement about what to do with a match; it replaces the minutes-to-hours of manual line comparison with a ranked shortlist produced in seconds, consistently, from any operator.

2. The two methods inside the application

Re-ID deliberately runs **two independent methods side by side** on every query, because they fail differently and agreement between them is itself information.

LOFAR-tonal (the familiar method, automated). The application extracts up to 20 narrowband tonal lines from the recording using a TPSW whitener — the same physics the analyst reads by eye — and scores candidates by how well line frequencies match. This column behaves the way an analyst expects a line-matching method to behave, including its weaknesses.

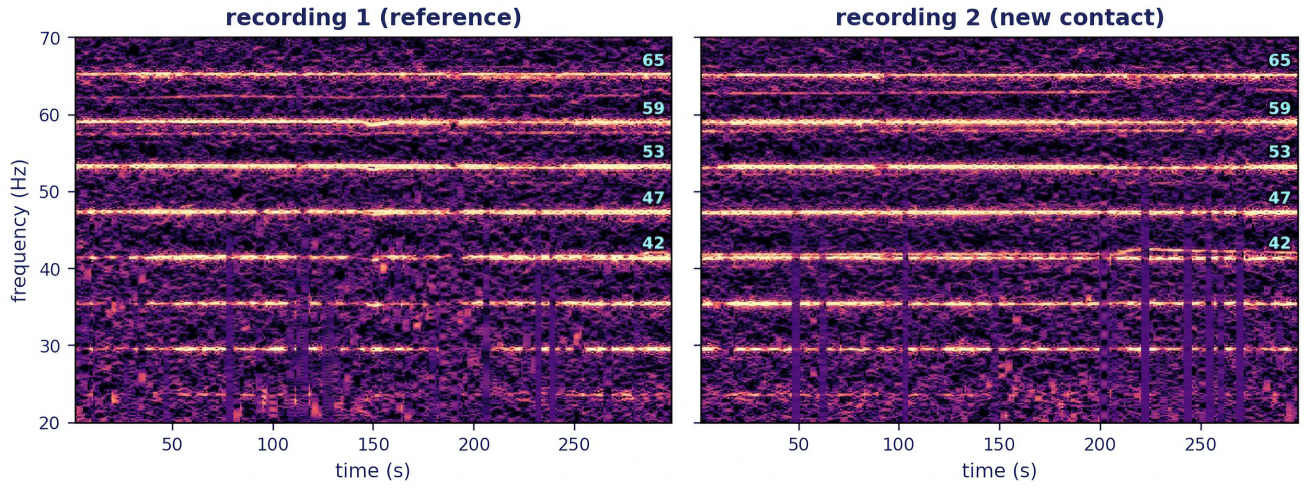


Figure 1: Two passages of the same vessel, tonal lines labelled — what the tonal method compares.

SKANN (the new method). A deep neural network reads the raw waveform and produces a 512-dimensional acoustic **fingerprint** of the vessel — a compact numerical signature capturing the full character of the radiated sound (harmonic structure, broadband texture, modulation), not just line positions. Two recordings are compared by the cosine similarity of their fingerprints.

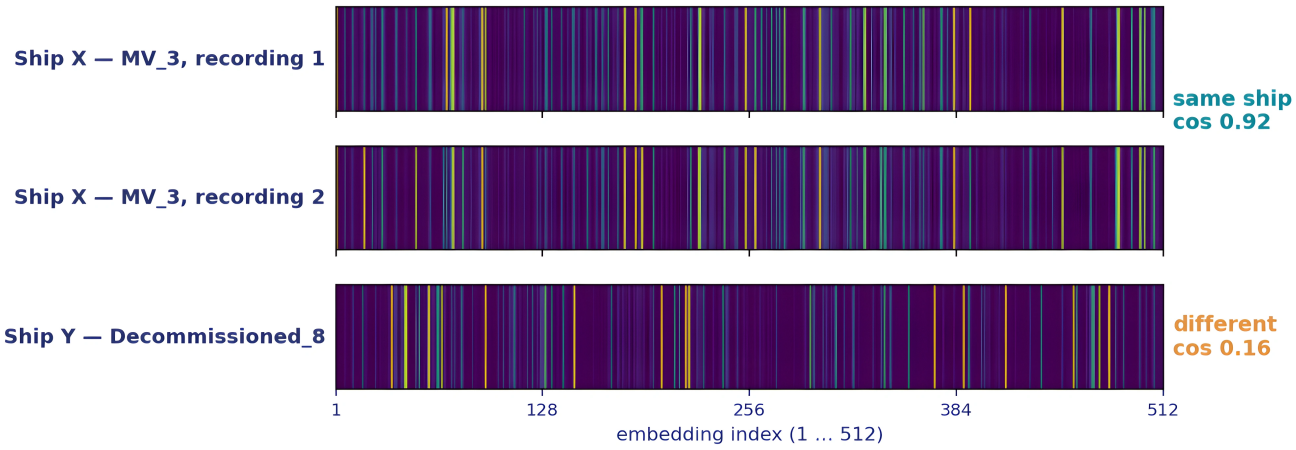


Figure 2: Waveform → network → 512-dimensional fingerprint — the enrol/query concept.

The two columns are never numerically fused. (We tested combining them into one score. It made results worse when the contact's speed changed between the two recordings.) When both methods rank the same vessel in the top 3, the app tags the agreement — that convergence of two unrelated mechanisms is the strongest single indicator the application offers.

3. Why SKANN — what it is designed to withstand

A ±4% change in a vessel's speed shifts every tonal line by ±4% of its frequency — enough to break frequency matching — and sea-state noise buries the weaker lines. The SKANN fingerprint does not rest on matching individual line frequencies: it captures the full character of the radiated sound (§2), and it is trained with augmentation that perturbs recording conditions (§5), so that recordings of the same hull stay close together. **A closing or opening contact in weather is the realistic case.**

4. What this means operationally

For the **operator**, Re-ID turns a specialist comparison task into a load-file-read-list task: a query takes seconds, needs no line-reading skill, and produces the same answer regardless of who runs it. For the **analyst**, it is a force multiplier and a second opinion — the shortlist arrives pre-ranked with both methods' views and the full evidence (scores and the tonal-line record, exportable as CSV) one click away, leaving the analyst's time for the judgement calls the machine cannot make: whether a 0.68 against a three-passage gallery entry in heavy weather is a call to escalate. For the **unit**, the gallery itself becomes an asset: every enrolled passage compounds, building an acoustic reference library that survives postings and shift changes.

The model is **frozen** — it never retrains in the field, so its behaviour is fixed, testable and certifiable. What grows is the **gallery**. And because the model has never been trained on the end user's own recording chain, performance on the end user's data is expected to improve further if the model is ever fine-tuned on it — an option, not a requirement, and gated by a formal evaluation.

5. A brief note on the technology (SKANN)

SKANN is a raw-waveform convolutional neural network (~4.6 M parameters) purpose-built for ship-radiated noise — there is no fixed spectrogram front end; the network learns its own time–frequency analysis. Audio is resampled to 8 kHz mono, cut into 5-second segments and z-normalised, then processed in three stages:

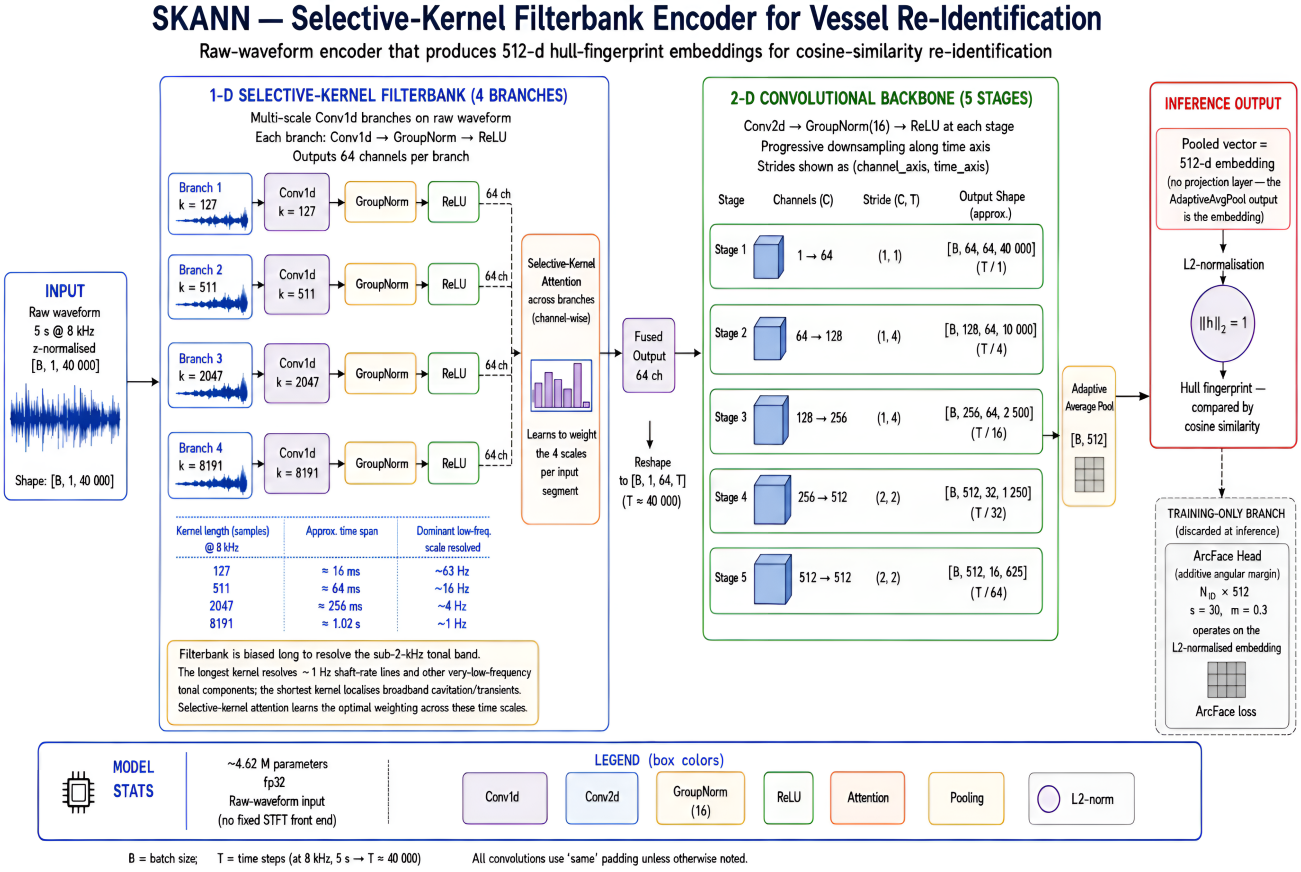


Figure 3: Model structure — SK filterbank → selective-kernel fusion → 2-D encoder → 512-d fingerprint.

SK Filterbank. Four parallel banks of learned filters run over the raw waveform at kernel lengths of 127 / 511 / 2,047 / 8,191 samples — at 8 kHz, analysis windows from ~16 ms to ~1 second. Short kernels localise broadband transients and cavitation in time; the longest kernels resolve frequency to ~1 Hz, sharp enough for slow shaft-rate lines. A **selective-kernel attention** stage then learns, per segment, how to weight and fuse the four scales — the network decides what matters in each recording rather than having frequency bands hand-assigned.

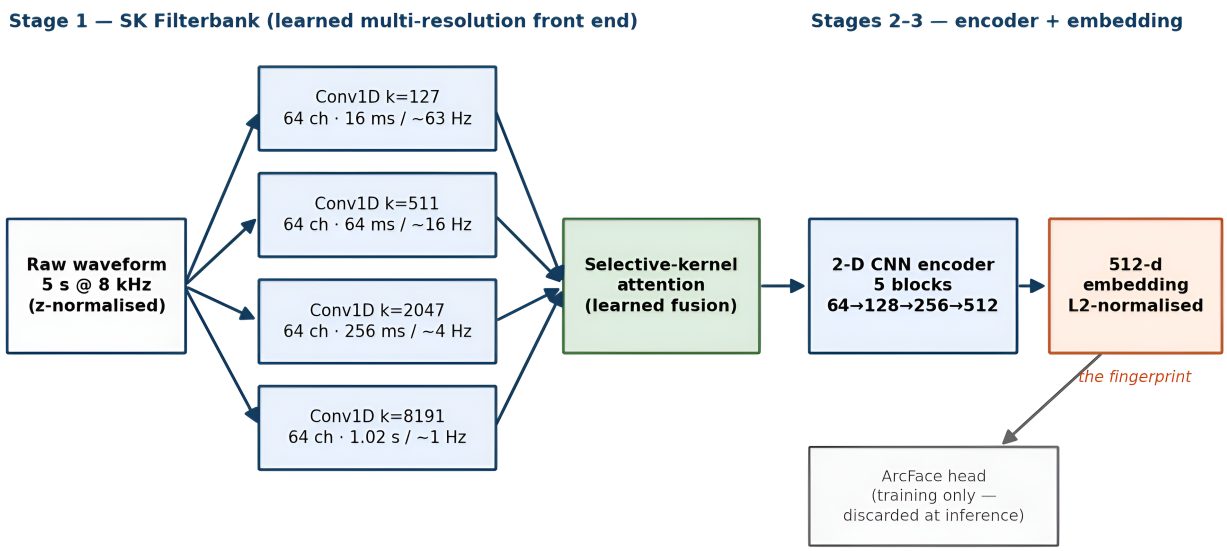


Figure 4: The same chain as a labelled schematic — kernel lengths, resolutions, and the training-only ArcFace head.

Convolutional encoder. The fused feature map is treated as a learned time–frequency image and passed through a five-block 2-D convolutional encoder, pooled to a single vector, projected to a **512-dimensional embedding** and normalised to unit length. That normalised vector *is* the fingerprint; a recording's fingerprint is the mean of its segment embeddings, re-normalised. Unit normalisation is what makes cosine similarity the correct comparison.

Training. The network was trained with an **ArcFace** angular-margin objective over hundreds of vessel identities drawn from multiple ocean datasets and recording systems, with heavy augmentation — added sea-state noise and speed perturbation among five transforms — so that recordings of the *same* hull cluster tightly on the unit hypersphere while *different* hulls are pushed apart by a margin, regardless of recording conditions. The ArcFace classification head exists **only during training**; at inference it is discarded and only the embedding backbone ships. The fielded model is therefore a fixed function from audio to fingerprint with no vessel list baked in — the vessel list is the gallery, and the gallery is yours.

Full training details — datasets, augmentation design, epoch-set construction, checkpoint lineage and evaluation protocol — are documented separately and are not part of this repository.

Part II — Installation and accounts

6. Requirements

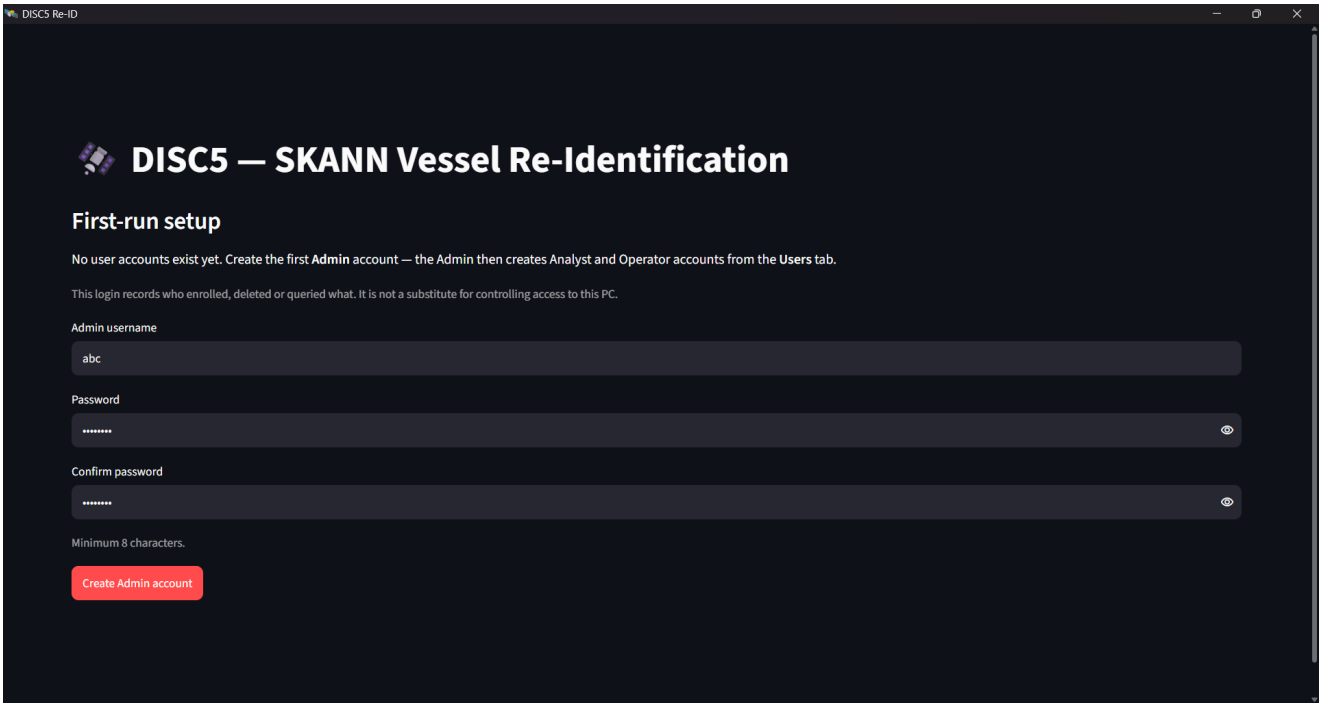
The machine. Windows 10/11 x64, an NVIDIA GPU (the application runs on CPU but embedding is impractically slow — minutes per segment; the sidebar must read `Compute: GPU`), ~6 GB free disk for the installed bundle, and a **writable** installation folder (not `C:\Program Files\`). The reference configuration is the operator workstation: Intel i9, 32 GB RAM, NVIDIA RTX 5060 Ti, Windows 11. The **only system prerequisite is a current NVIDIA graphics driver** — verify with `nvidia-smi` in PowerShell, which must print a table naming the card. No Python, no internet, and no other software is required — the bundle is self-contained, including the CUDA libraries (cu130 build, carrying the Blackwell kernels the RTX 50-series needs) and VC++ runtime DLLs.

Input files. WAV format only. Any sample rate (the application resamples to 8 kHz internally); mono or stereo (stereo is mixed down automatically); any bit depth soundfile can read (16/24/32-bit PCM or float). Minimum usable length is ~5 seconds after resampling — shorter clips are rejected with a message. There is no upper length limit, but the upload control caps single files at 200 MB; for large files or bulk enrolment, use the **folder** mode (§10), which reads directly from disk with no size cap.

The clock. Activity-log timestamps come from the machine's own clock. An air-gapped PC has no time synchronisation — verify the system date and time at installation and periodically after.

7. Installation and first run

1. Obtain `Re-ID.zip` from the v1.3 release page of the `disc5-reid` repository — https://github.com/suniltyagialtair/disc5-reid/releases/tag/v1.3 (access is by repository invitation) — and verify its integrity: `CertUtil -hashfile Re-ID.zip SHA256` must match the value in `SHA256SUMS.txt` published alongside it.
2. Extract to a writable folder, e.g. `C:\DISC5\Re-ID\`. The folder then contains `disc5_gui.exe` and `_internal\`.
3. Run `disc5_gui.exe`. If Windows shows a blue "Windows protected your PC" (SmartScreen) message, click **More info** → **Run anyway** — expected for an in-house, unsigned application on an offline machine; if antivirus quarantines files, add the Re-ID folder to its exclusions and re-extract. The application opens directly in its own desktop window — no browser is launched, none needs to be installed, and no console window appears. The interface is reachable **only from this machine**, by design. **Closing the window is how you shut the application down at the end of the session.**
4. **First-run setup.** On a fresh installation the application asks you to create the first **Admin** account before anything else.



Screenshot 01: first-run setup — creating the first Admin account.

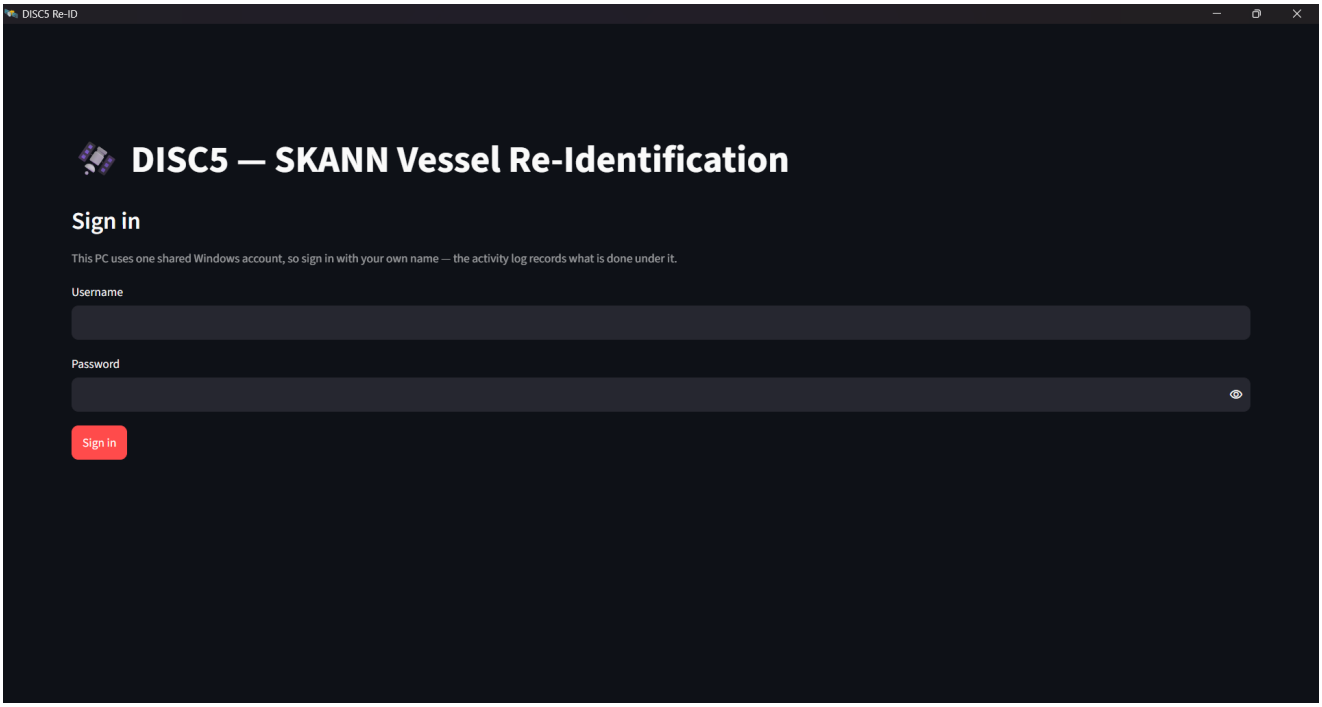
Choose the Admin username and a password of at least 8 characters. This account governs all others; record the credentials per your unit's procedure. If the Admin password is ever lost, recovery is possible — see the *Re-ID Administrator Security Note* (DISC5_ReID_Admin_Security_Note_protected.pdf). Password-protected; ask the maintainers for the password.

First demonstration (recommended). `DISC5_demo_clips_v2.zip` (~0.6 GB) is attached to the v1.2 release page — https://github.com/suniltyagialtair/disc5-reid/releases/tag/v1.2 — a demonstration set of 24 vessels: a *gallery* folder (one reference recording per vessel, with the metadata sidecar) and a *query* folder (two later passages per vessel). After creating accounts, extract it (e.g. `C:\DISC5\demo_clips\`), enrol the gallery folder from the Enrol tab's folder mode (§10) — MMSI and IMO fill in automatically from the sidecar — then identify recordings from the query folder (§11). The correct vessel should appear at or near the top of the SKANN column, usually with the ✓ agreement tag. This walks the entire workflow end-to-end — and it teaches the enrolment-first protocol on clips where the right answer is known. Treat it as a workflow demonstration, not a measure of accuracy on your own recordings.

8. Accounts and roles

Sign-in is required for every session. Three roles, strictly nested — each includes everything below it:

Role	Can additionally
Operator	Identify recordings, view ranked results, read the Gallery tab, export query-scoped CSVs
Analyst	Enrol recordings; delete passages or vessels; clear the gallery; gallery-wide exports
Admin	Manage user accounts (Users tab); view and export the activity log (Activity tab)

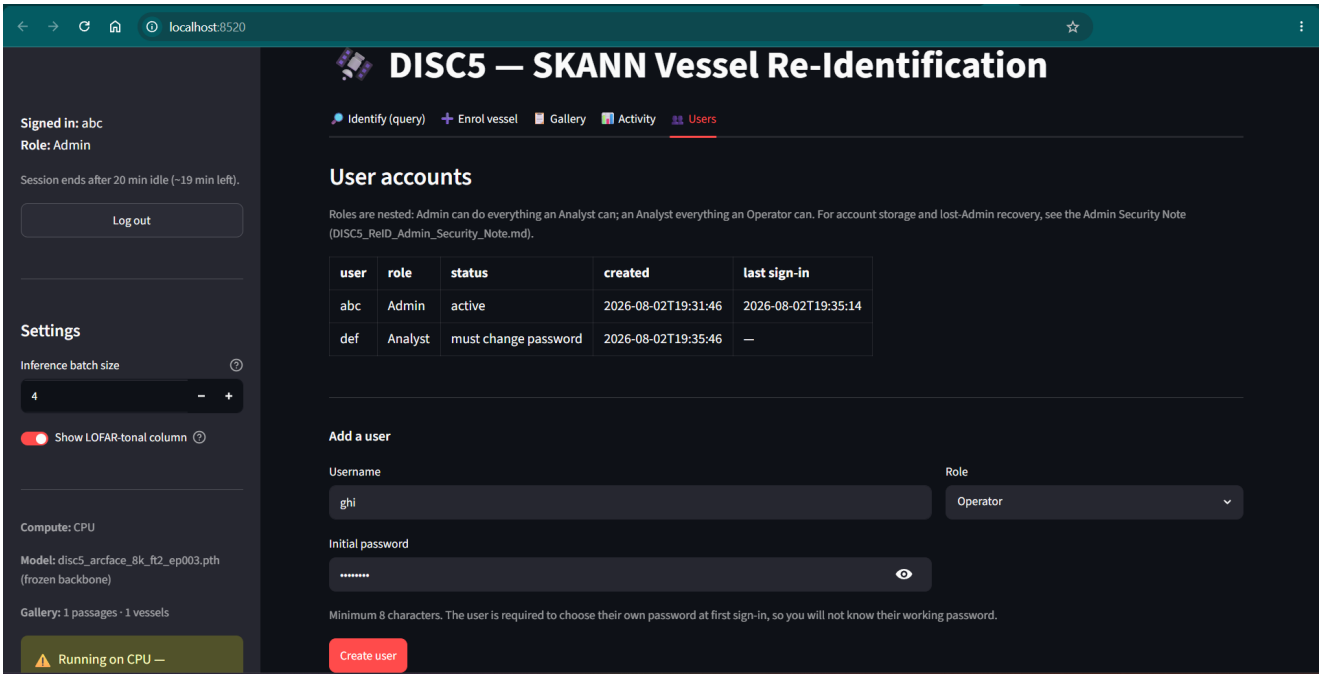


Screenshot 02: the sign-in screen.

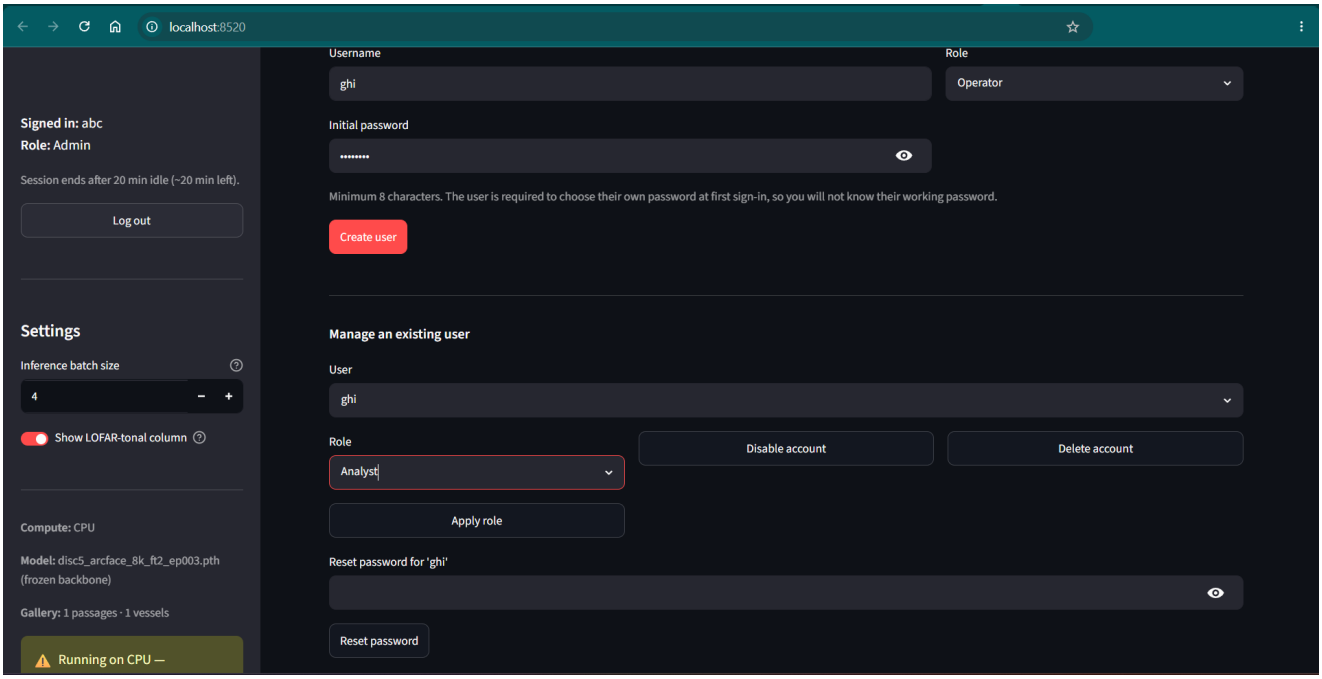
What the sign-in is for. The sign-in puts a name against every action and keeps destructive functions away from accidental use by restricting them to Analysts. Passwords are stored hashed, so no user — including the Admin — can read another's password. The security model, its limits, and credential recovery are covered in the *Re-ID Administrator Security Note*, which the administrator should hold.

9. Onboarding users (Admin)

1. Sign in as Admin (§8).
2. Open the **Users** tab.
3. Under **Add a user**: username (use people's names, not role words — the activity log records the username), role, and an initial password. The new user is required to set their **own** password at first sign-in, so the initial password is a one-time handover secret.
4. The same tab manages existing accounts: change role, disable/enable, delete, reset password (which again forces a change at next sign-in).



Screenshot 03: the Users tab — accounts table and the "Add a user" form.



Screenshot 04: managing an existing user — here the Admin is changing 'ghi' from Operator to Analyst; disable, delete and password-reset controls alongside.

The application refuses to demote, disable or delete the **last enabled Admin** — user administration can never be locked out by accident. Sessions end after 20 minutes of inactivity; five consecutive failed sign-ins lock an account for 5 minutes (both events are logged).

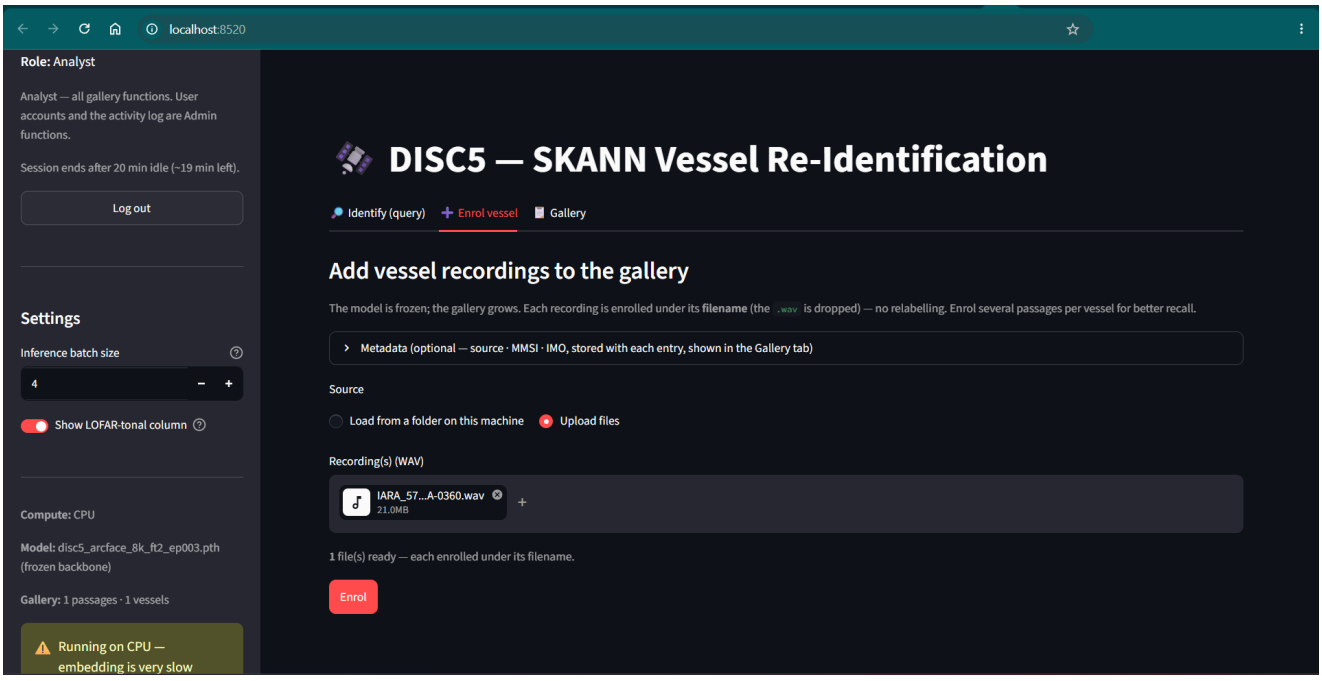
Password policy (stated defaults): minimum 8 characters; no complexity rules; no expiry. Complexity and expiry requirements, if mandated later, are configuration changes, not redesigns. The reasoning behind these defaults is in the *Re-ID Administrator Security Note*.

Part III — Operating the application

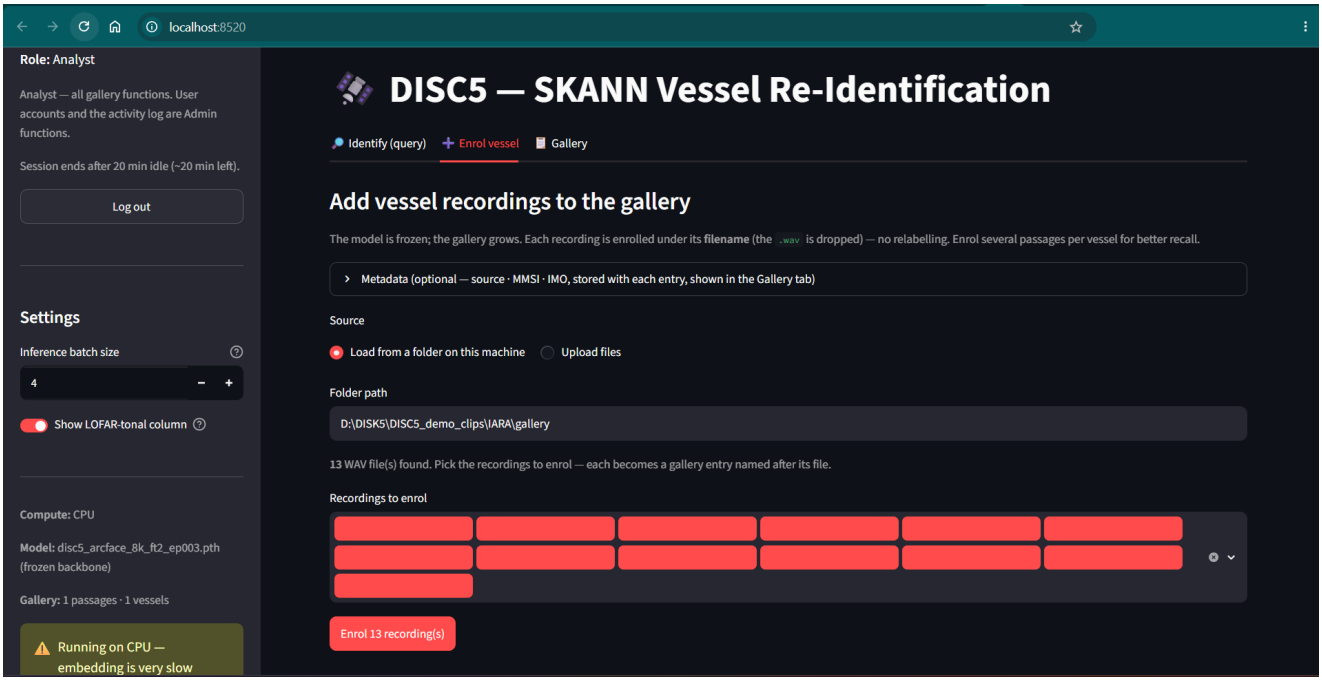
10. Enrolling vessels (Analyst)

Enrolment is how the gallery — the reference library every query is compared against — is built. Each enrolled recording becomes one **passage**: its 512-d SKANN fingerprint plus its top-20 tonal lines, stored under the recording's filename. A vessel may (and should) hold several passages: **enrolling multiple passages per vessel is the single biggest lever on recall**, because a query scores against every passage and reports the best per vessel.

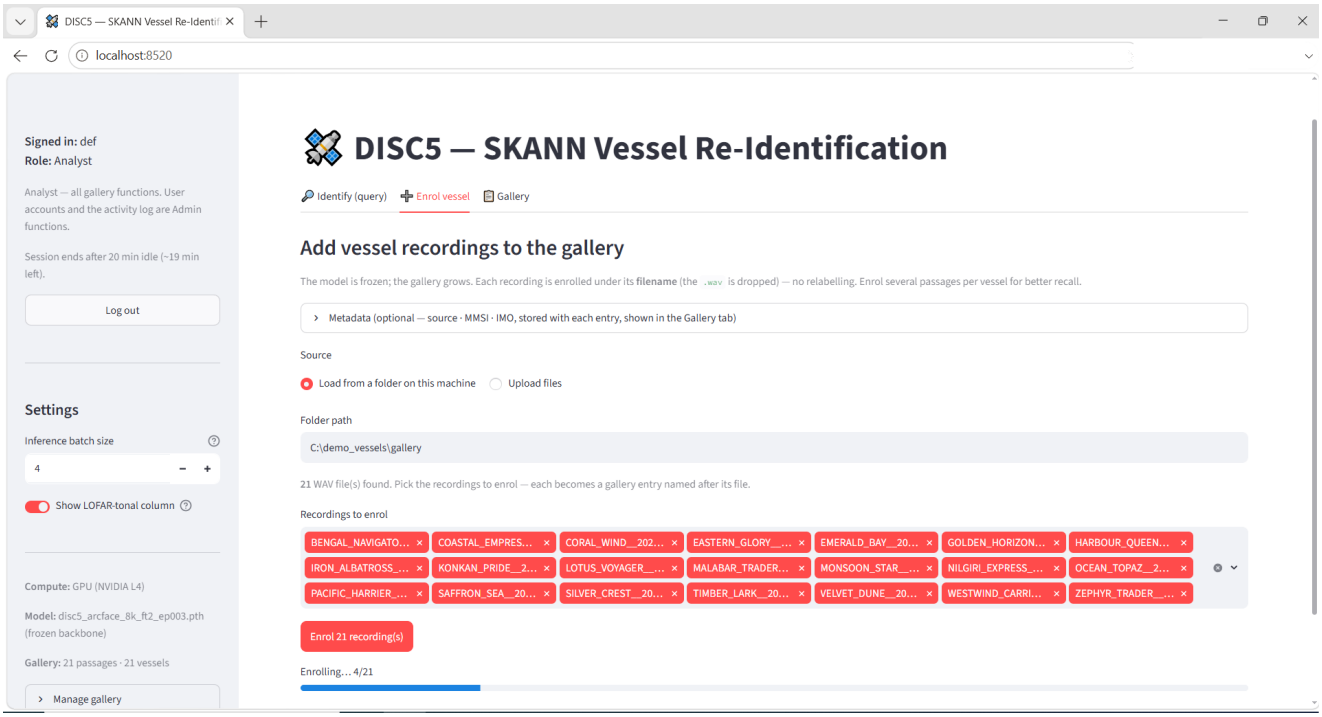
1. Open the ➕ **Enrol vessel** tab.
2. Optionally set metadata — source, MMSI, IMO. Left blank, source is auto-detected from the filename prefix (`ONC_`, `IARA_`, `DC_`, `SHIPSEAR_`) and MMSI and IMO from a `*clip_map.csv` sidecar if present (columns `vessel_mmsi`/`mmsi` and `vessel_imo`/`imo`); anything typed overrides.
3. Choose the input mode: **Load from a folder** (path on this machine; no file-size limit; best for bulk) or **Upload files** (drag-and-drop; 200 MB/file cap). Select the recordings and press **Enrol**.
4. Each recording is processed in turn — progress is shown; too-short clips are skipped with a message, not fatal.



Screenshot 05: the Enrol tab — a single recording uploaded, ready to enrol under its filename.



Screenshot 05b: folder mode — 13 demo recordings selected for one enrolment run.



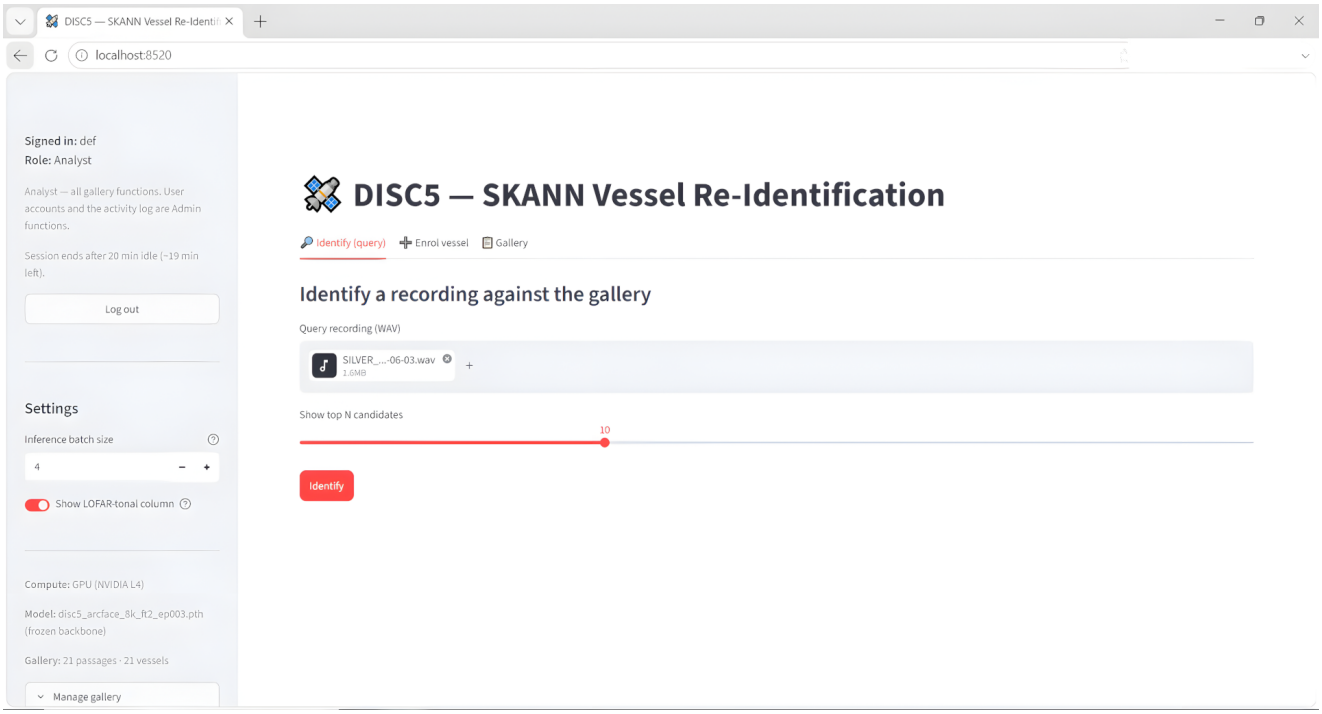
Screenshot 06: bulk enrolment in progress — 21 recordings selected from folder mode, 4 of 21 processed.

> Vessel names, dates, MMSI and IMO numbers appearing in Screenshots 06–11 are illustrative demonstration data, not real vessels. The demonstration set — including its changed-speed queries (Screenshot 08) — is prepared demonstration material for walking the workflow.

Naming matters: the filename becomes the gallery label (the `.wav` extension is dropped). `MV_ALPHA__passage1.wav` is a useful label; `rec(7).wav` is not. There is no relabelling — delete and re-enrol to rename. Good practice: enrol clean, representative passages, several per vessel. Mixing sources in one gallery (end-user, IARA and ONC recordings together) is fine — but a sparse or very different-sounding gallery can make the top rank look more confident than it is.

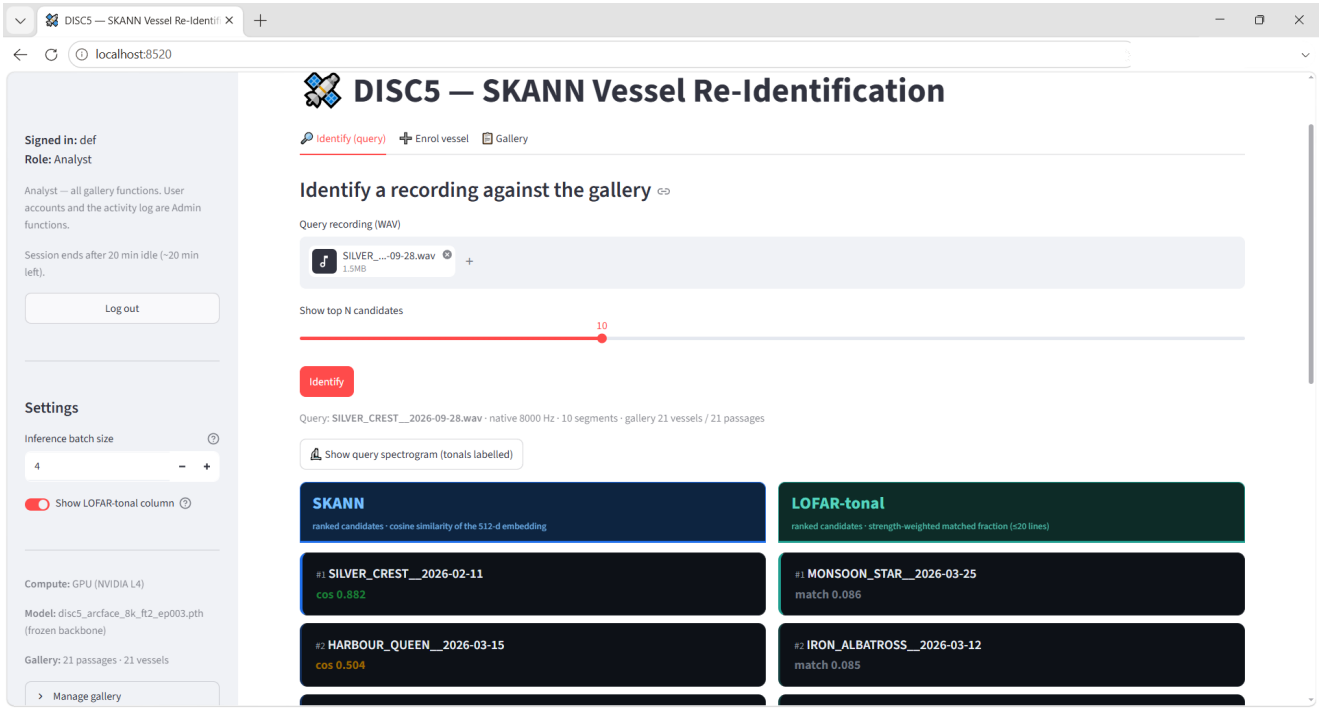
11. Identifying a recording (all roles)

1. Open the 🔎 **Identify (query)** tab, upload the query WAV, choose how many candidates to display, press **Identify**.



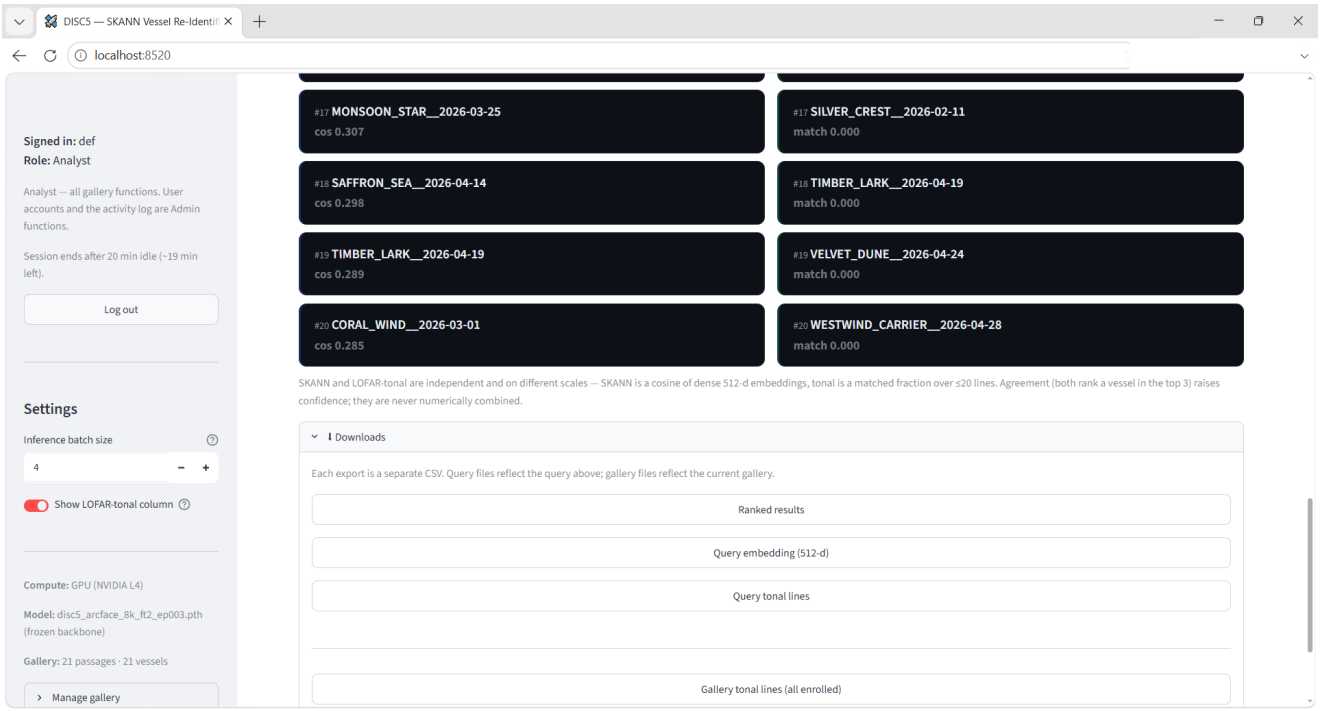
Screenshot 07: the Identify tab — query recording loaded, top-N set to 10, ready to identify against the demonstration gallery.

2. Embedding takes seconds on GPU. The result is two ranked lists side by side — SKANN (blue) and LOFAR-tonal (teal).



Screenshot 08: ranked results — a changed-speed repeat query of an enrolled vessel against the demonstration gallery. SKANN: rank 1 correct at cosine 0.882, with rank 2 far behind at 0.504. LOFAR-tonal:

the correct passage scores 0.000 (rank 17); the column's best is a wrong vessel at 0.086. §12a walks through this result.



Screenshot 08b: the same result scrolled to the tail of the lists — the correct passage sitting at rank 17 in the tonal column at match 0.000 — and the Downloads panel beneath, where the ranked results, the query fingerprint and the tonal-line CSVs are exported.

The enrolment-first protocol — the single most important operating rule. The application can only match against what is enrolled. Querying a vessel that is **not** in the gallery correctly returns low-scoring wrong candidates — that is the system working, not failing. A meaningful test of the system is: enrol one recording of a vessel, then query a **different** recording of the same vessel. Likewise in operations: a flat, low-scoring result is the expected signature of an out-of-gallery contact.

12. Reading the results

Each candidate card shows rank, vessel label and score. The colour is a rough display band, **not a decision rule**:

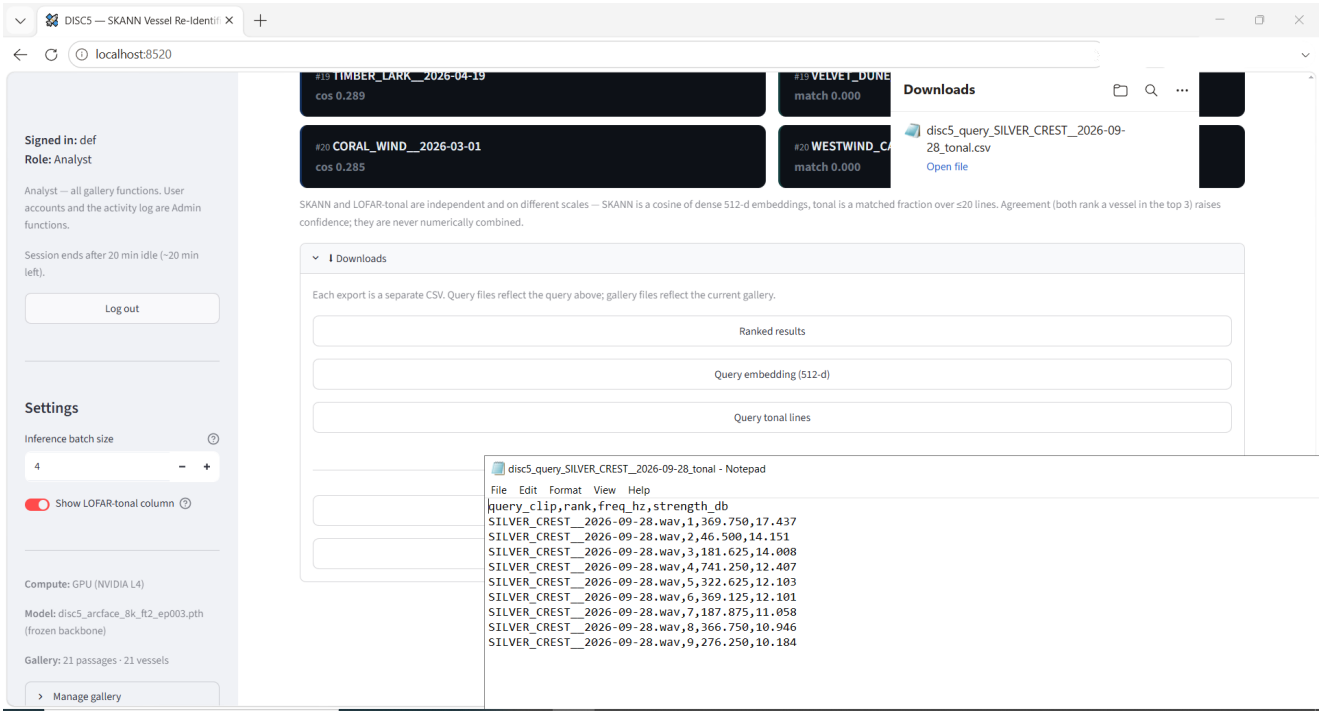
	Green	Amber	Grey
SKANN cosine	≥ 0.65	0.45–0.65	< 0.45
Tonal match	≥ 0.35	0.15–0.35	< 0.15

The two scales are **not comparable to each other** — SKANN is a cosine over dense fingerprints, tonal is a strength-weighted matched fraction over ≤ 20 lines. Compare each column against its own ranking. What to weigh, in order: **agreement** (both methods top-3 on the same vessel — the strongest indicator, tagged ✓); **margin** (a rank-1 well clear of rank-2 means more than its absolute value); **conditions** (in noise or with a manoeuvring contact, expect lower absolute scores — §3 — and lean harder on agreement and margin); and **gallery depth** (a match against a three-passage vessel is worth more than against a single passage).

Re-ID is a shortlisting aid, not an automatic identification. On unseen vessels across genuinely different encounters, the correct vessel is the single top match less than half the time — but is usually within the top few.

Review the shortlist rather than acting on rank 1 alone, and confirm with the tonal second opinion, the tones CSVs, and your own analysis.

Evaluating the tonals — use the CSVs. The way to verify the tonal evidence line by line is to download the **Query tones CSV** (§13; the Downloads panel in Screenshot 08b): up to 20 detected lines with their frequency in Hz and strength in dB — the exact lines the tonal score uses. An Analyst can additionally download the candidate's **Gallery tones CSV** and compare the two frequency lists directly; matching lines within a few Hz are the tonal method's evidence for the match. (The spectrogram view is disabled in this version; its button points to these same CSV downloads.)



Screenshot 09: the downloaded Query tones CSV opened for inspection — one row per detected line, strongest first. This particular query is the changed-speed passage of Screenshot 08: comparing its frequencies against the vessel's enrolled lines shows every line displaced by a common factor — the CSV diagnosing the speed change that broke the tonal score.

12a. Worked example — a contact at changed speed

§3 explains why a speed change breaks tonal matching while SKANN is designed to hold. Screenshot 08 (§11) is exactly such a query, live from the demonstration set: a repeat passage of the same vessel at a changed speed. SKANN: rank 1 correct at cosine 0.882 — versus 0.891 on the vessel's steady-speed passage, a loss of 0.009. LOFAR-tonal: the correct passage falls from 0.328 on the clean run to 0.000 (rank 17), with a wrong vessel topping the column at 0.086 — one speed change took it to zero, exactly the §3 mechanism.

A flat, near-zero tonal column under a manoeuvring contact is therefore **expected behaviour**, not a fault — read the SKANN column and the margin, and use the tones CSV (Screenshot 09) to confirm the uniform frequency shift that explains the tonal collapse.

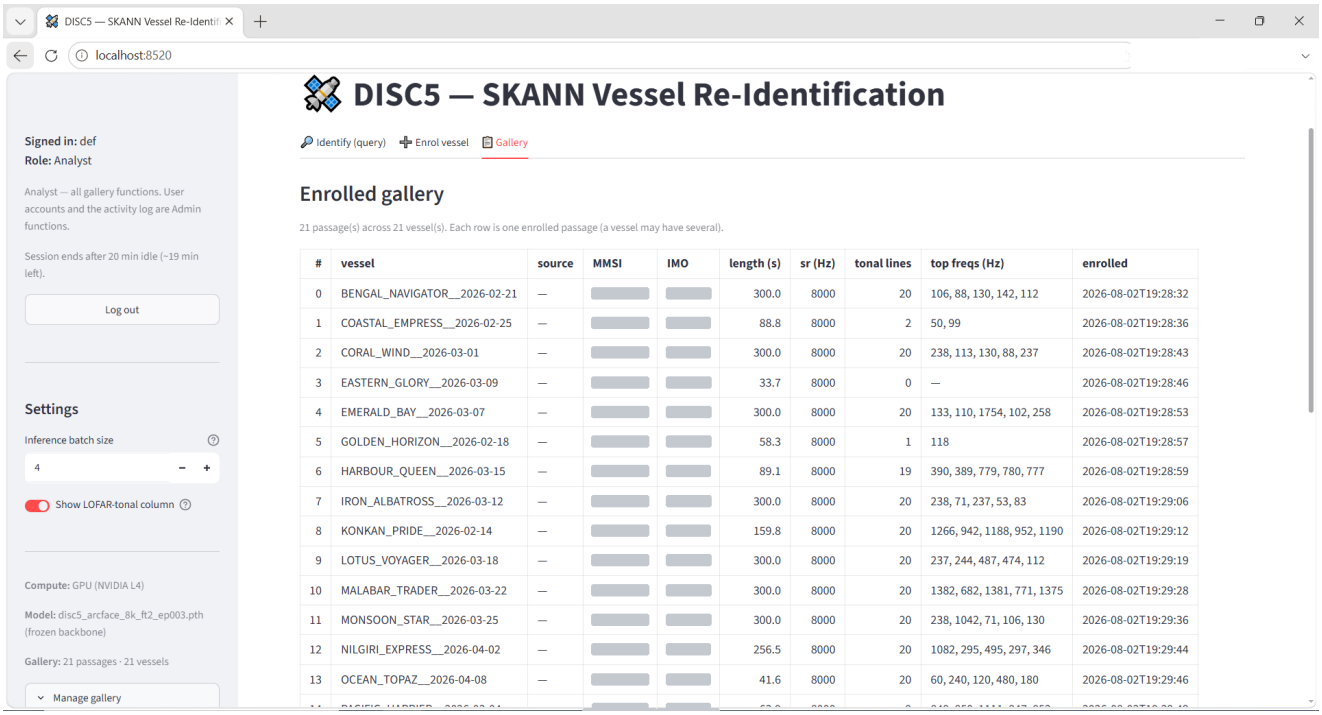
13. Exports

Each download button opens a standard Windows save dialog asking where to save the file. Every export is logged.

Download	Format	Contents	Who can
Ranked results	CSV	`query_clip`, `native_sr`, `n_segments`, `rank`, `skann_candidate`, `skann_cos`, `tonal_candidate`, `tonal_match`	everyone
Query fingerprint	CSV	`query_clip`, `dim`, `value` — 512 rows	everyone
Query tones	CSV	`query_clip`, `rank`, `freq_hz`, `strength_db` — up to 20 rows	everyone
Gallery tones	CSV	`candidate`, `passage_key`, `rank`, `freq_hz`, `strength_db`	Analyst
Gallery tones (wide)	CSV	one row per enrolled recording: details plus all 20 tone pairs	Analyst
Gallery fingerprints	CSV	`candidate`, `passage_idx`, `dim`, `value`	Analyst
Activity log	CSV	the recorded actions	Admin

14. The Gallery tab

Read-only for Operators: every enrolled passage with its vessel, source, MMSI/IMO, clip length, sample rate, line count and top frequencies — the reference for interpreting a result.



#	vessel	source	MMSI	IMO	length (s)	sr (Hz)	tonal lines	top freqs (Hz)	enrolled
0	BENGAL_NAVIGATOR__2026-02-21	—			300.0	8000	20	106, 88, 130, 142, 112	2026-08-02T19:28:32
1	COASTAL_EMPRESS__2026-02-25	—			88.8	8000	2	50, 99	2026-08-02T19:28:36
2	CORAL_WIND__2026-03-01	—			300.0	8000	20	238, 113, 130, 88, 237	2026-08-02T19:28:43
3	EASTERN_GLORY__2026-03-09	—			33.7	8000	0	—	2026-08-02T19:28:46
4	EMERALD_BAY__2026-03-07	—			300.0	8000	20	133, 110, 1754, 102, 258	2026-08-02T19:28:53
5	GOLDEN_HORIZON__2026-02-18	—			58.3	8000	1	118	2026-08-02T19:28:57
6	HARBOUR_QUEEN__2026-03-15	—			89.1	8000	19	390, 389, 779, 780, 777	2026-08-02T19:28:59
7	IRON_ALBATROSS__2026-03-12	—			300.0	8000	20	238, 71, 237, 53, 83	2026-08-02T19:29:06
8	KONKAN_PRIDE__2026-02-14	—			159.8	8000	20	1266, 942, 1188, 952, 1190	2026-08-02T19:29:12
9	LOTUS_VOYAGER__2026-03-18	—			300.0	8000	20	237, 244, 487, 474, 112	2026-08-02T19:29:19
10	MALABAR_TRADER__2026-03-22	—			300.0	8000	20	1382, 682, 1381, 771, 1375	2026-08-02T19:29:28
11	MONSOON_STAR__2026-03-25	—			300.0	8000	20	238, 1042, 71, 106, 130	2026-08-02T19:29:36
12	NILGIRI_EXPRESS__2026-04-02	—			256.5	8000	20	1082, 295, 495, 297, 346	2026-08-02T19:29:44
13	OCEAN_TOPAZ__2026-04-08	—			41.6	8000	20	60, 240, 120, 480, 180	2026-08-02T19:29:46

Screenshot 10: the Gallery tab — one enrolled passage per vessel, MMSI and IMO auto-filled from the sidecar file. Note the vessel with 0 detected tonal lines (a quiet 33.7 s clip): SKANN enrols and matches it even where the tonal comparator has nothing to work with.

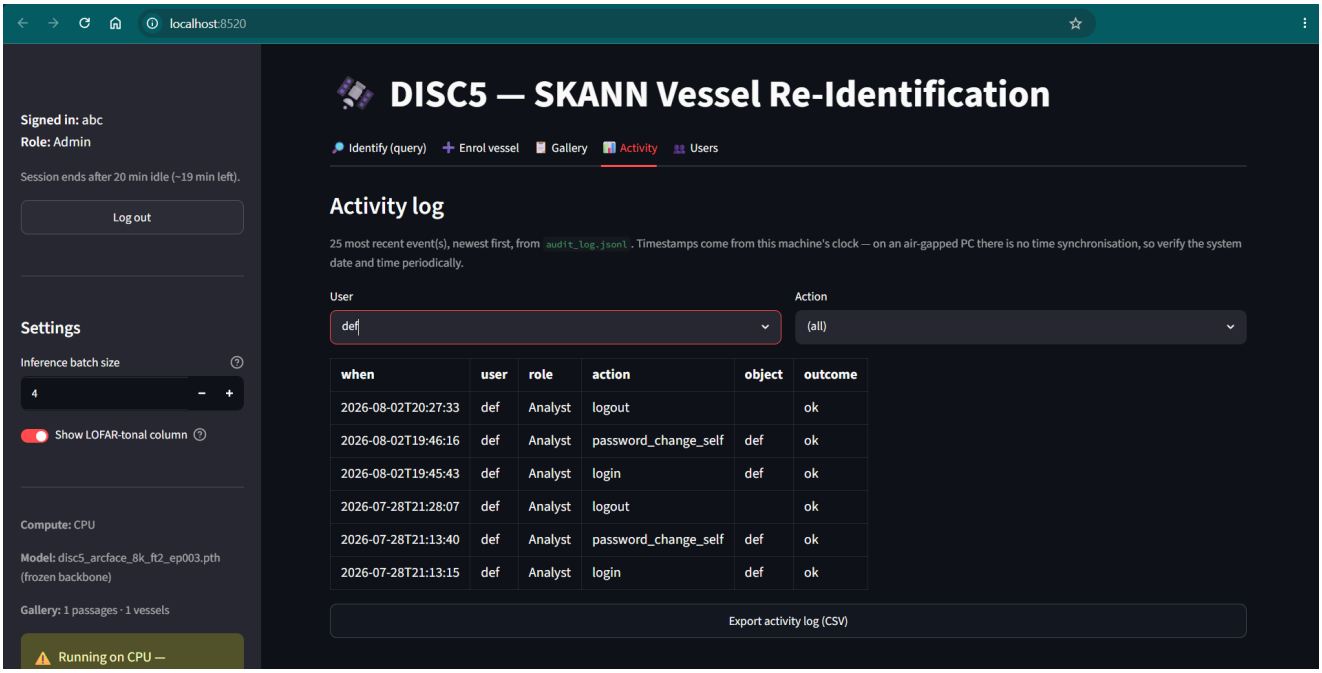
Analysts additionally: delete any single passage (removed from both stores together), remove a whole vessel or clear the gallery (sidebar → Manage gallery), and the wide CSV exports.

Gallery size does not slow queries down: comparing a query against the whole gallery is a single fast calculation, so the time a query takes is set by processing the query itself, not by how many vessels are enrolled.

15. The activity log (Admin)

Everything of consequence is recorded: sign-ins (including failures and lockouts), account changes, enrolments, deletions, queries (with segment count and top match), exports, sign-outs and session timeouts.

In the application: the **Activity** tab shows the most recent events newest-first, filterable by user and action, exportable as CSV.



Screenshot 11: the Activity tab with a filter applied.

On disk: `audit_log.jsonl` beside the executable — append-only, one JSON object per line:

```
{"action":"query","host_account":"Admin","object":"DC_D9_S3.wav","outcome":"ok",
 "role":"Operator","ts":"2026-07-28T16:09:44","user":"Operator1","n_seg":
12,"top_match":"DC_D9_S1"}
```

Reading it: `ts` local machine time; `user`/`role` the signed-in identity; `action`/`object` what was done to what; `outcome` `ok` or a failure reason (`fail_password`, `fail_no_user`, `lockout`); trailing fields per-action detail. `host_account` is the Windows account — on a shared-login machine it is constant and carries no information; the `user` field is the accountable identity.

16. Upgrading to a new version

Before extracting a new `Re-ID.zip` over an existing installation, **preserve the four state files** beside `disc5_gui.exe`: `users.json` (accounts), `audit_log.jsonl` (activity history), `gallery.npz` (fingerprints), `gallery_tonal.json` (tonal lines). Copy them aside, extract the new bundle, copy them back. Skipping this loses every account and every enrolled vessel. Verify the new zip's SHA-256 against its `SHA256SUMS.txt` before extracting.

17. Troubleshooting

Symptom	Cause and remedy
Sidebar shows `Compute: CPU`	Graphics driver not active. Run `nvidia-smi`; if it fails, install a current NVIDIA driver and reboot, then relaunch
Windows blocked the app ("protected your PC")	SmartScreen on an unsigned in-house app: **More info → Run anyway**; add the folder to antivirus exclusions if files were removed
Window did not open	Check `disc5_desktop_log.txt` beside `disc5_gui.exe` for the reason; if the log shows the server started but no window appeared, re-extract the bundle (the rendering component may have been quarantined — add the folder to antivirus exclusions)
Enrolment "succeeds" but the gallery stays empty	Installation folder is not writable — move out of `C:\Program Files\`
"Query too short"	Clip under ~5 s after resampling to 8 kHz
Upload rejected at 200 MB	Per-file upload cap; use folder mode (§10)
Account locked	5 failed sign-ins → 5-minute lockout; wait, or an Admin resets the password
Admin password lost	Recoverable — see the *Re-ID Administrator Security Note* (DISC5_ReID_Admin_Security_Note_protected.pdf; password-protected; ask the maintainers for the password); gallery and activity log are unaffected
App unreachable from another machine	By design — the interface binds to `localhost` only
How do I uninstall?	There is no uninstaller — delete the Re-ID folder (copy out `gallery.npz`, `gallery_tonal.json`, `users.json`, `audit_log.jsonl` first if they must be kept) and, optionally, the display cache at `%LOCALAPPDATA%\DISC5_ReID\`

Appendix

Fixed signal settings

8 kHz mono; 5 s / 40,000-sample segments; per-segment z-normalisation; TPSW tonal whitener (window 8 Hz, guard 1.5 Hz, α 3.0); a tone is detected only if it holds more than 8 dB above the background for at least 10 seconds, between 3.5 and 2000 Hz; ≤ 20 tonal lines per recording; 512-d L2-normalised embeddings, mean-pooled per recording. These mirror the training pipeline exactly and are not operator-adjustable — changing them would silently desynchronise scores from the validated benchmarks.

Honest-use notes for demonstrations

Describe fine-tuning on the end user's data as **"expected to improve further"**, never as a guarantee. Brief the enrolment-first protocol (§11) before any hands-on session — un-briefed testing of un-enrolled vessels is the known path to a false "it returned the wrong ship" impression.

Screenshot capture checklist

Screenshots 01–11 are embedded above (08/08b = the changed-speed query demonstration and its scrolled tail; 09 = its Query tones CSV opened for inspection). 01–05b and 11 were captured on the local build (Chrome); 06–10 on the packaged GPU build (Edge), at 100% browser zoom, full window, with browser account and assistant buttons painted out of the toolbar. Screenshots 06–10 use illustrative demonstration vessel data throughout. In 05b the vessel names on the selection chips are blanked, and in 10 the MMSI and IMO values are blanked. For future captures: 100% zoom, full window, packaged app (not dev mode); use test vessel names, not operational ones, for any copy that leaves the site. v1.3: screenshots showing the first-run and sign-in screens (01, 02) were recaptured in the native application window so the window frame and title bar match what operators see; in-tab screenshots are unaffected in content and may be reshot opportunistically.